Considering a Bacterial Model for Ecological Nanoparticle Toxicity

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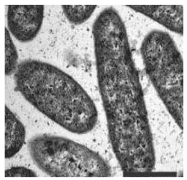
Abstract

Nanomaterials are notable for their extremely small size as well as their unique physical and chemical properties. Nanomaterial behavior in the environment is dictated by surface properties such as surface charge and hydrophobicity. Recent increase in use of engineered nanomaterials in biomedical, electronic and consumer products provides the motivation for assessing and understanding the toxicity of released engineered nanoparticles on the ecosystem. Due to the important roles bacteria play in the ecosystem and global nutrient cycling, understanding nanoparticle impact on low trophic level organisms like bacteria is critical. Motivated by the current lack of understanding of nanoparticle interactions with bacteria, this study evaluates gold nanoparticle interactions with *Shewanella oneidensis* MR-1 by measuring changes in bacterial respiration upon nanoparticle exposure. Gold nanoparticles have been chosen due to their common use. Through this research, gold nanoparticle concentrations and surface properties detrimental to bacteria will be identified, giving critical insight into the design of environmentally friendly nanoparticles.

Introduction

- Lux Research predicted nano-enabled products to grow by $3.1 trillion by 2015 (Lux Research).
- Among the many materials being utilized by this growing field is gold nanoparticles (Au NPs).
- Au NPs are used in electronic conductors, photodynamic therapy, drug delivery, sensors, probes for biological imaging, in vitro diagnostics, and catalysis.
- Engineered nanoparticles like gold can be released into the environment where they may interact with plants, microbes, and other organisms.
- Investigation of nanoparticle fate in natural environments is therefore necessary to ensure a healthy ecosystem.

Model System: *Shewanella oneidensis* MR-1



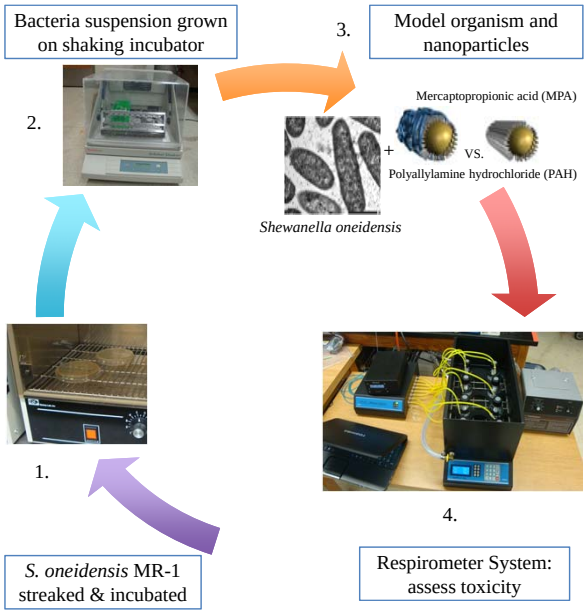
S. oneidensis

Characteristics:
- Genus distributed throughout the world in a diverse environments
- Gram-negative, metal reducing bacterium
- Can live under aerobic or anaerobic conditions

Significance of Study

- Understand how nanoparticles impact biological systems from a molecular perspective for environmentally sustainable solution.
- Use Respirometer System to asses the respiration of a single cell model – *Shewanella oneidensis* MR-1 – following nanoparticle exposure with respect to particle surface functionality

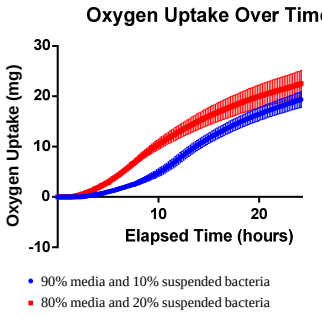
Methods



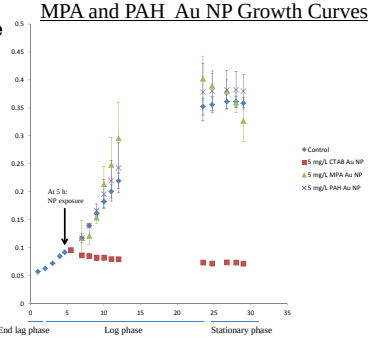
Future Research

- Conduct more respirometer experiments assessing response of *Shewanella oneidensis* interaction with varying gold nanoparticle surface functionality (PAH and MPA)
- Evaluation of other nanoparticles such as silver, nanodiamond, CdSe/ZnS quantum dots, and titanium dioxide utilizing the respirometer system

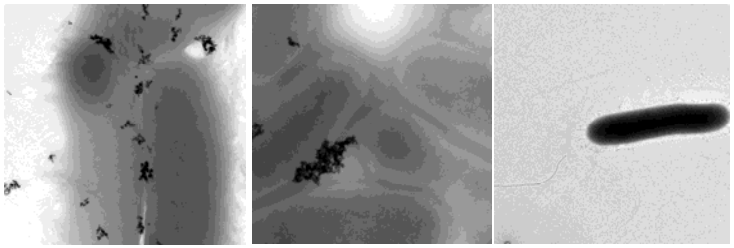
Results



MPA and PAH Au NP Growth Curves



Transmission Electron Microscopy (TEM) Images of Au NPs with Cells



PAH Au NP	MPA Au NP	Negative Control
8 mg/L PAH Au NP with 10^8 cells/mL *Shewanella* in minimal media	5 mg/L MPA Au NPs with 10^8 cells/mL *Shewanella* in minimal media	10^8 cells/mL *Shewanella* in minimal media without Au NPs

Conclusion

- Higher sample volume (decreased head space) increases sensitivity and lowers variation in the Respirometer data
- Growth curves show no significance effect of MPA and PAH Au NPs on *Shewanella* growth rate
- TEM images of Shewanella with MPA or PAH Au NPs:
  - PAH appears to induce some morphology changes
  - MPA interacts with cells to a lesser degree

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